Statistics and Data Analysis

Week 2

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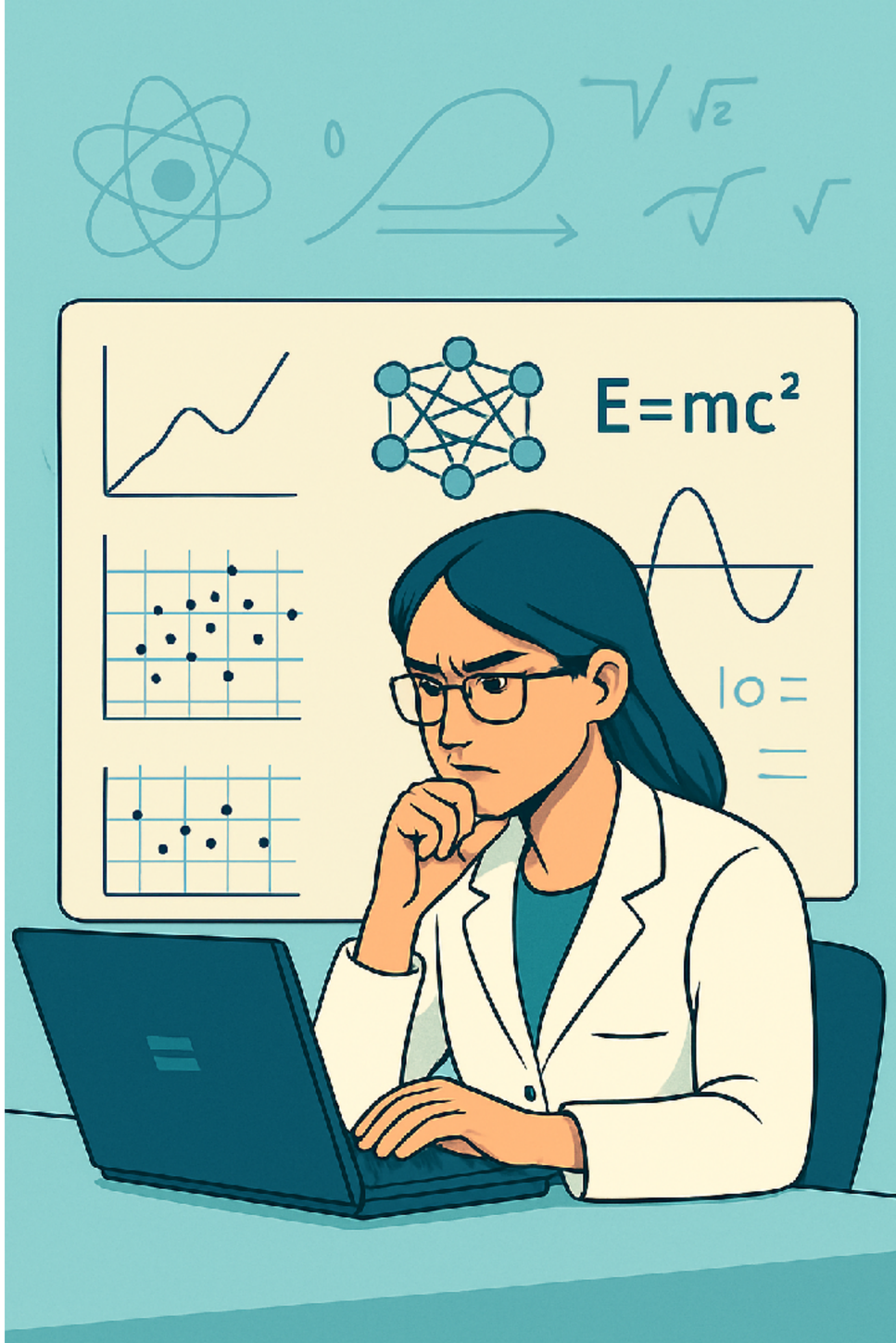


Solutions to last week’s exercise

Week1_Exercise_Solutions.ipynb

Program of the first 4 weeks (March)

Days	Thu 12:45 - 14:30 (2h)	Fri 13:45 - 15:30 (2h)
05/03 06/03	- Probability: foundations and interpretations - Practice with PDFs, CDFs, Quantiles - Joint PDFs, moments, correlation, transformations	- Practical CLT and the lognormal - Linear error propagation - MC methods - **Exercise**
12/03 13/03	- Frequentist hypothesis testing - Optimal tests, Neyman-Pearson, p-value - Estimators and properties	- Likelihood - MLE - **Exercise**
19/03 20/03	- Asymptotic MLE properties and MVB - Advanced MLE applications and MC methods - Frequentist Confidence intervals	- Frequenstist vs Bayesian upper limit - **Exercise**
26/03 27/03	- Bayesian priors and posterior construction - Profile likelihood and Wilks' theorem - Bayesian parameter estimation	- Bayesian model selection and evidence - Marginal likelihood, Bayes factor - **Exercise**

Frequentist hypothesis testing

Main reference: G.Cowan chapter 4

The hypothesis

A hypothesis specifies the probability of the data x

- x could be the outcome of an observation, continuous/ discrete, univariate/multivariate
- Doing an hypothesis is equivalent to writing $x \sim P(x \mid H)$

N.B.: in a frequentist sense there is no $P(H)$ so that is not strictly speaking a conditional probability

- Simple hypothesis: $P(x \mid H)$ completely specified
- Composite hypothesis: (or family of hypotheses) when H contains unspecified parameters

For *composite hypotheses* we will define $P(x \mid H)$ as the likelihood function $L(H)$, a function of the fixed data that depends on the parameter(s) of H

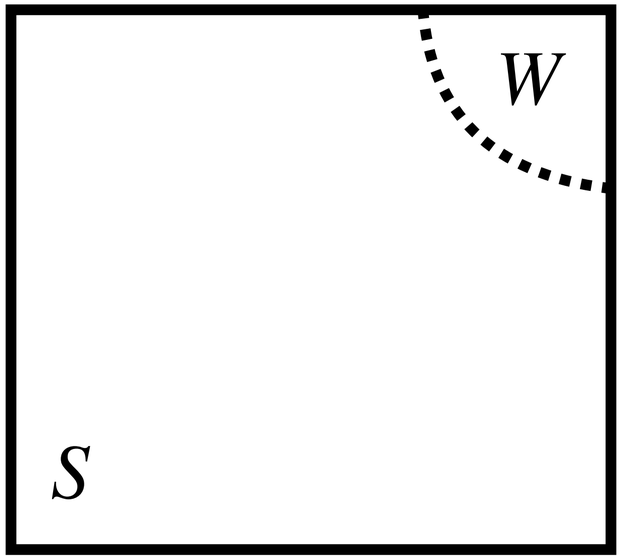
The hypothesis test

- **Goal:** make statement about H based on observed data x

We define a test of the hypothesis H_0 by specifying a **critical region** W of the sample space S such that there is (no more than) a probability α to observe the data there **assuming** H_0 **is correct**

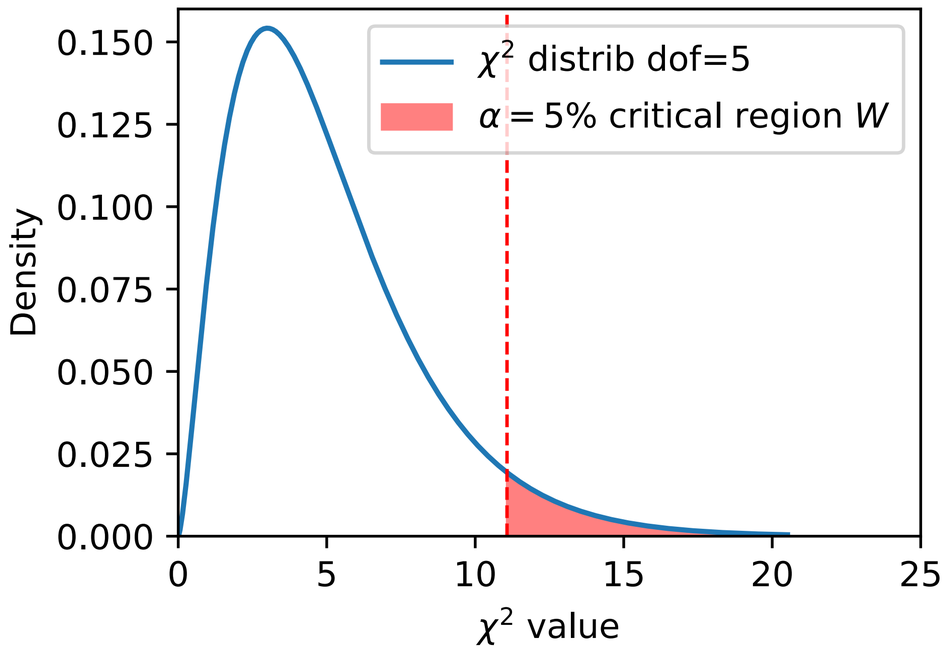
$$P(x \in W \,|\, H_0) \leq \alpha$$

- α is called the size of the test and is equal to some small value, e.g. 5%
- If the data x is observed in the critical region → reject H_0



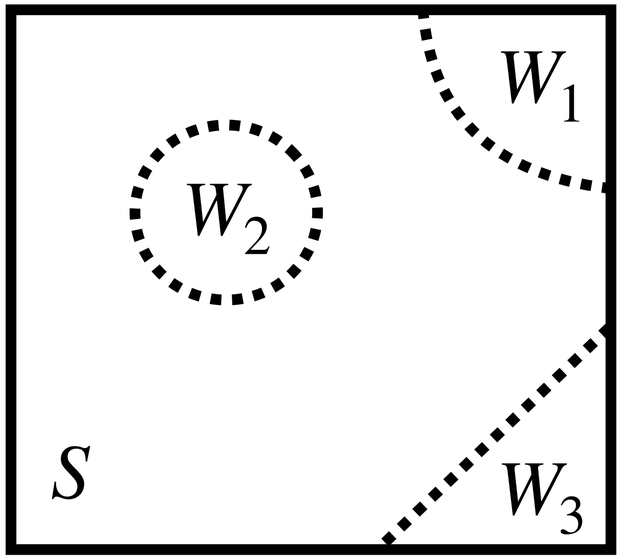
Example: test goodness of linear fit

- You should be familiar with the χ^2 test for goodness of fit
- Classic example:
  - You do a linear fit using least squares to N data point
  - The linear fit has two variables. The fit has $N - M$ d.o.f.
  - You compare the χ^2 value obtained with the distribution expected for the hypothesis of a good fit (i.e. the χ^2 distribution for 5 degrees of freedom)
  - You define the critical region W with size $\alpha = 5\,\%$ for large χ^2 values that would indicate a bad fit (reject H_0)



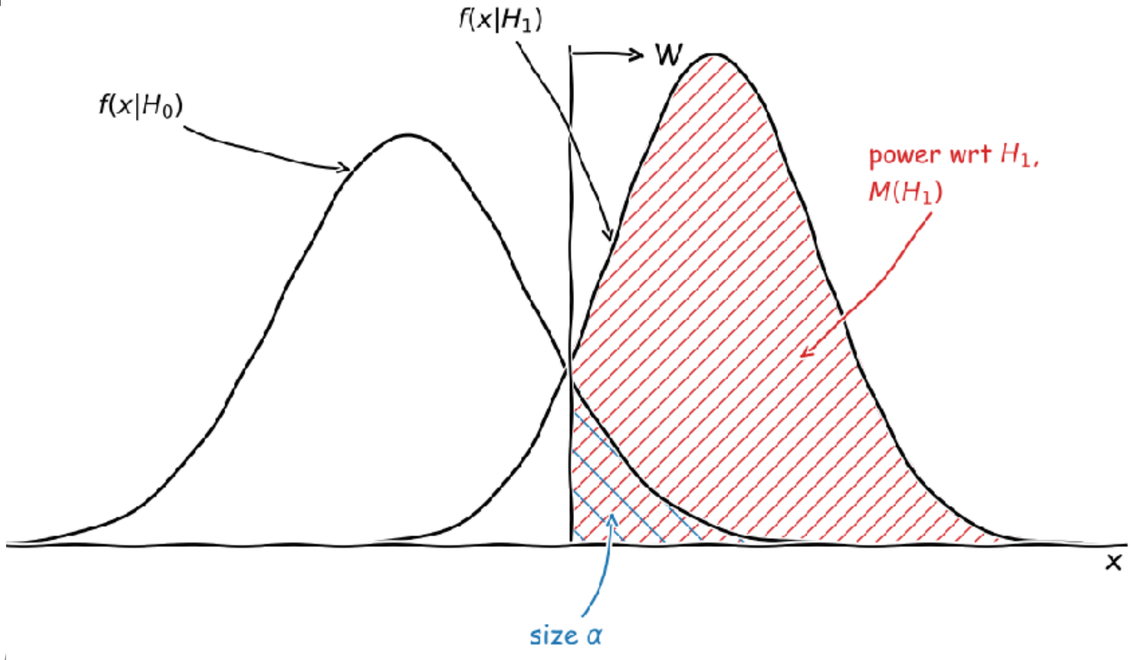
Choice of the critical region

- But in general there is an infinite number of possible critical region W_i with size α
- We need an alternative hypothesis H_1 to motivate where to place the critical region!
- A region of S that is “not characteristic” of H_0 is meaningful only if it is more characteristic of H_1
- (1930s discussion between Fisher and Neyman+Pearson)



The alternative hypothesis H_1

- Place the critical region where there is a low probability (α) to find data if H_0 is true, but a high probability $M(H_1)$ if H_1 is true
- The probability $M(H_1)$ is called **power of the test** with respect to H_1
- The alternative hypothesis could be a family of hypotheses consistent with that choice of critical region

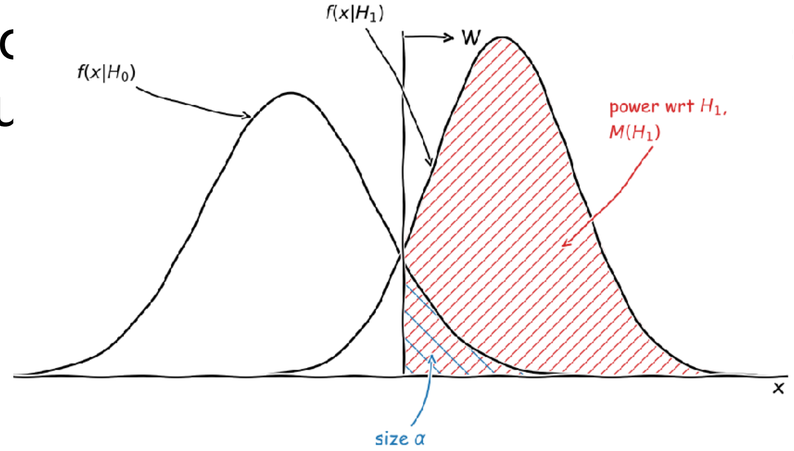


Meaning of a frequentist hypothesis test

- What does it mean to reject an hypothesis?
- In **frequentist statistics** we only associate probability with data (outcomes of repeatable observations)
- If you want to assign a degree of belief to a hypothesis you need to use a **Bayesian approach** and that requires you assign prior probabilities $\pi(H)$ to the hypotheses that are relevant to the problem:

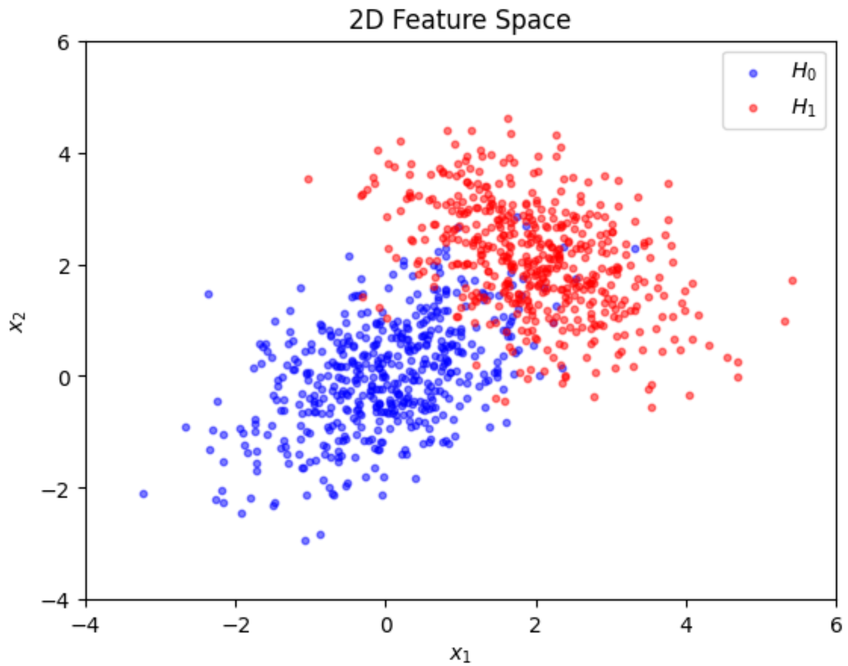
$$P(H|x) = \frac{P(x|H)\pi(H)}{\int P(x|H)\pi(H)dH}$$

- But then what is a frequentist hypothesis test?
  - if rejecting a hypothesis does not mean that you think it is false then what good is it?
  - A frequentist test can compute the probability to accept/reject a hypo or assu ue

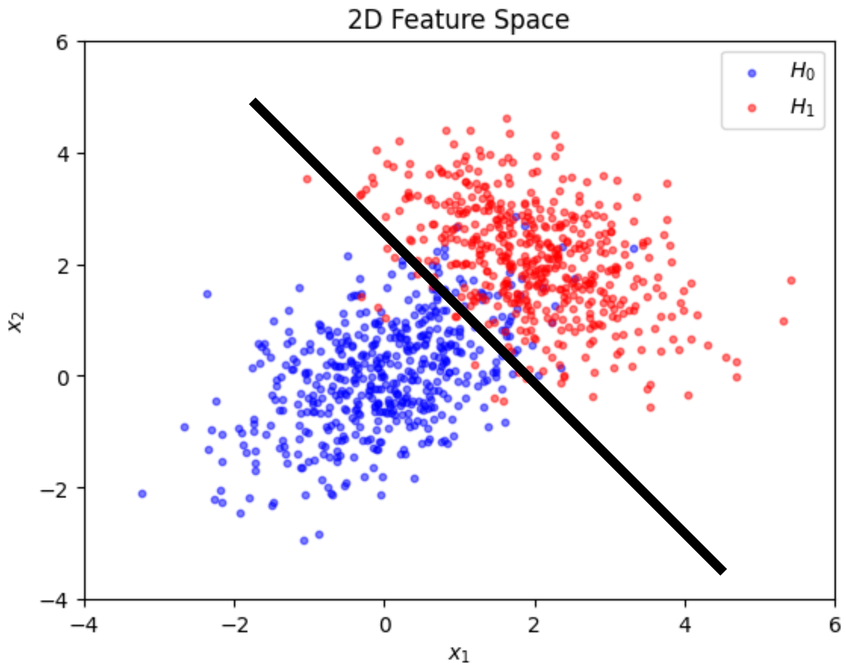


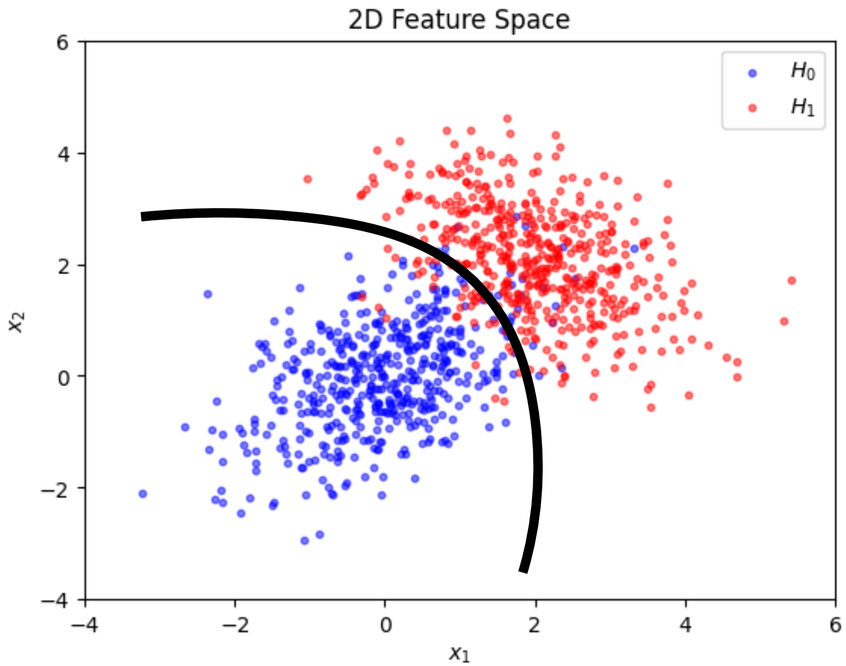
The test statistic

- How do we define the critical region in multiple dimensions? It lacks natural ordering
- Let's consider a dataset with two populations
  - we have measured two features x_1 and x_2 of the data and we want to test the hypotheses:
  - H_0: the data point is from the **blue** population
  - H_1: the data point is from the **red** population
- We want to define a critical region with a decision boundary
- N.B.: In an n-dimensional space, the boundary will be a hypersurface



Choosing the decision boundary



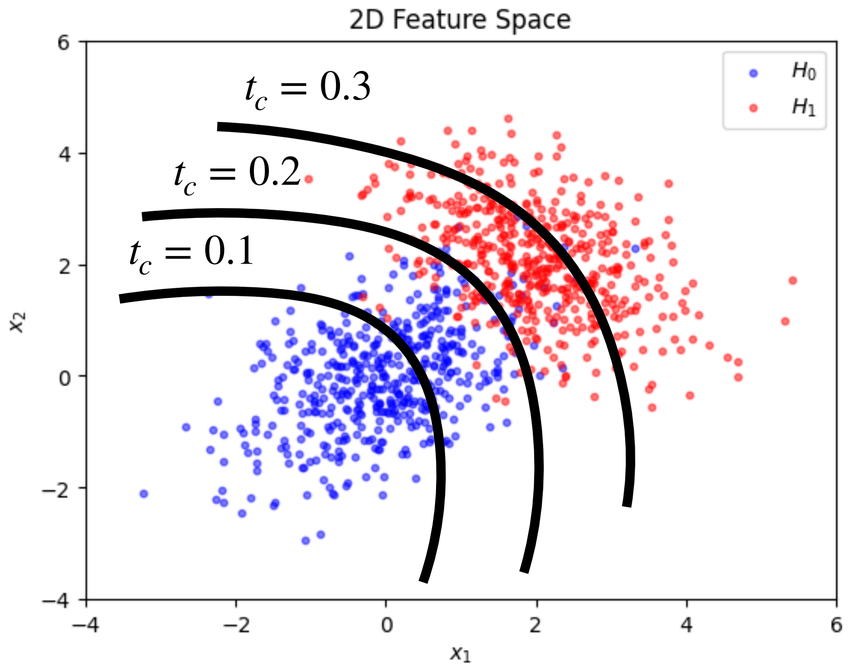


The test statistic

- Can define a hypersurface in n-dimension by a scalar function of the data:

$$t(x_1, \ldots, x_n) = t_c$$

- We call this the test statistic (or the decision function)
- Different values of t_c identify a family of surfaces
- Note that being a function of the data, the test statistic is also a random variable

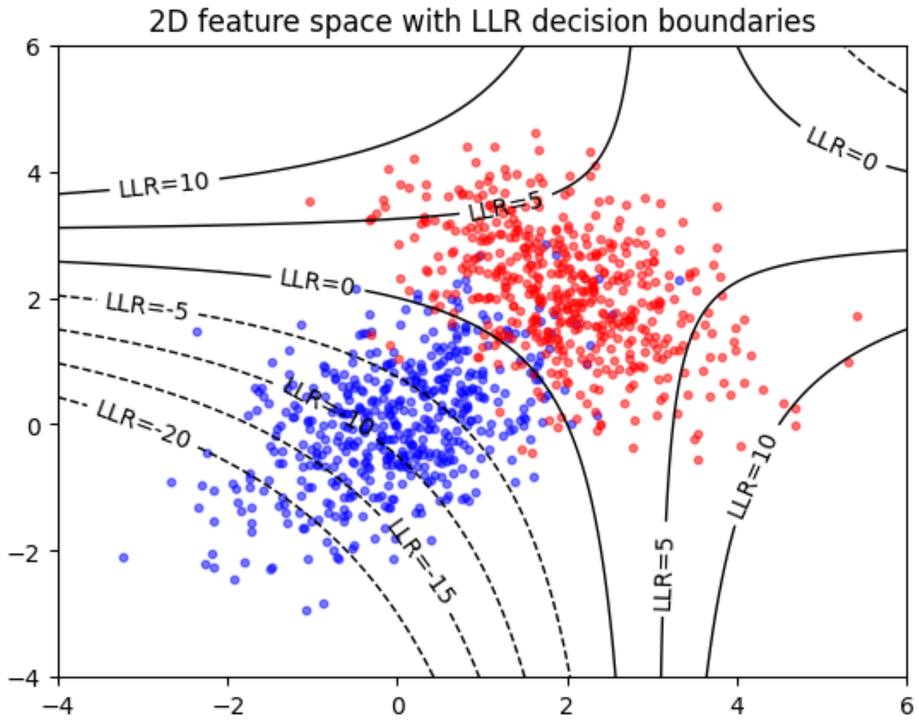


Optimal test statistic / decision boundary

- The Neyman-Pearson lemma
- *The likelihood ratio is the optimal test statistic*: for a given test of H_0 of size α it gives the highest power with respect to the alternative H_1

$$t(x) = \frac{P(x \mid H_1)}{P(x \mid H_0)}$$

- N.B. any monotonic function of $t(x)$ (for example the log) leads to the same test
- Demonstration of the NP lemma can be found in text books

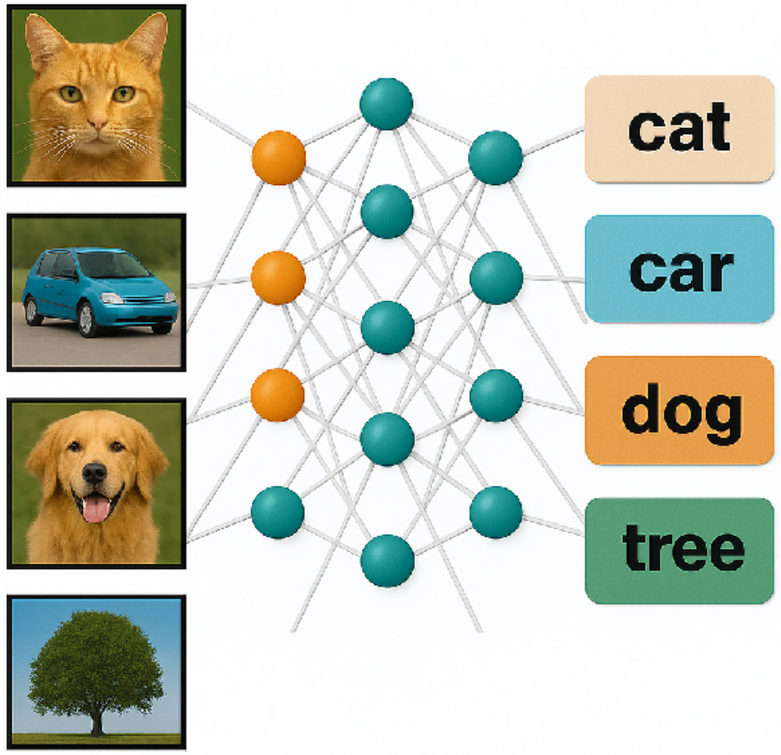


Jupyter notebook

Week2_NPLemma.ipynb

Why Machine Learning classifiers?

- If you know the PDFs of the two classes, then thanks to the NP lemma you already know the best possible classifier, it's simply the likelihood ratio!
- Why training Machine Learning classifiers then?
  - In real-world dataset we normally do not know the multivariate PDFs of the training datasets
  - Estimating a multivariate PDF in high dimensions (e.g., a 1000×1000 pixel image) is computationally and statistically nearly impossible (the curse of dimensionality)



Why don’t we have the PDFs?

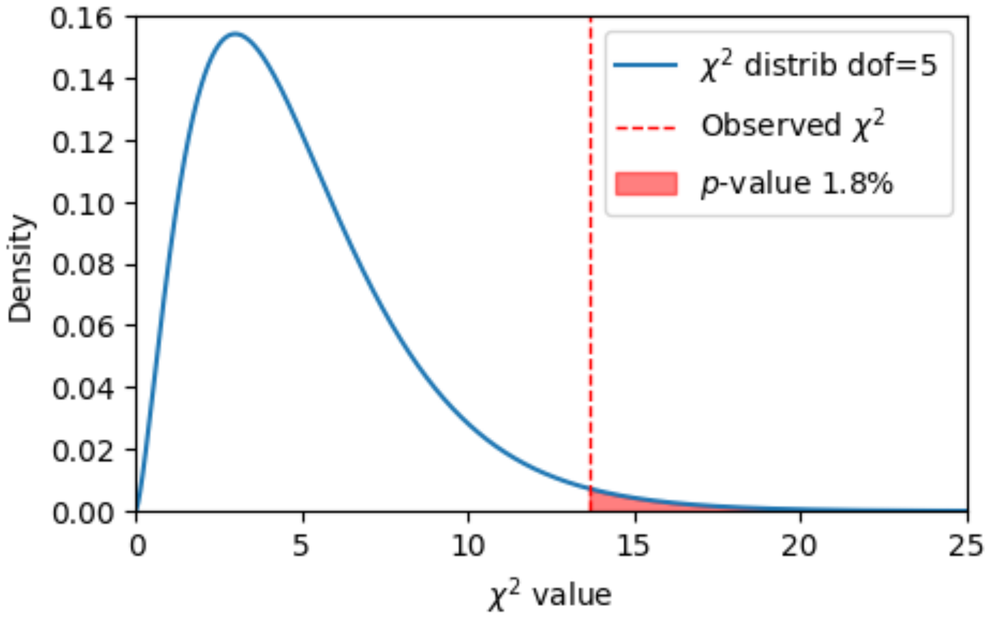
- Very often we don't have the explicit formulae for the PDFs of the different hypotheses
- Instead we often have MC models which we can use to produce simulated data
  - Often in MC simulation the hypothesis does not predict the PDF for the measured variable x, but rather for some intermediate set of latent variables z_1
  - After sampling $z_1 \sim f(z_1 \,|\, H)$ the MC simulation follows many other steps $z_2 \sim f(z_2 \,|\, z_1)$, $z_3 \sim f(z_3 \,|\, z_2)$, ...
- So even if H is defined and we can generate x according to it, the formula $f(x \,|\, H)$ is a very difficult (impossible) integral to compute:

$$f(\mathbf{x} \mid H) = \int \cdots \int d\mathbf{z}_1 \cdots d\mathbf{z}_n f\left(\mathbf{x} \mid \mathbf{z}_n\right) f\left(\mathbf{z}_n \mid \mathbf{z}_{n-1}\right) \cdots f\left(\mathbf{z}_2 \mid \mathbf{z}_1\right) f\left(\mathbf{z}_1 \mid H\right)$$

The *p*-value

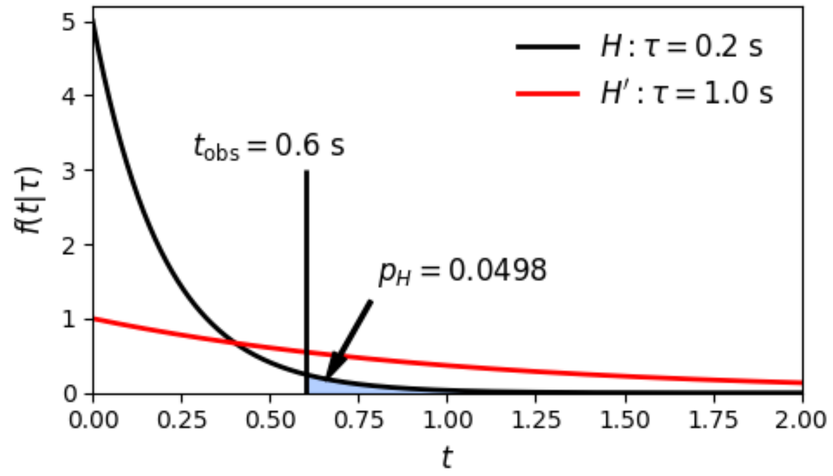
Definition of *p*-value

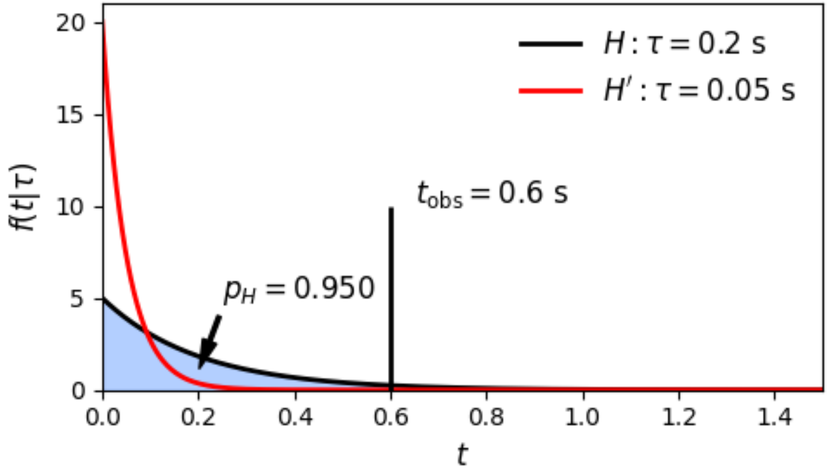
- Outcome of goodness-of-fit or significance test is the so-called p-value
- The p-value is defined as the probability under the assumption of H to observe data as discrepant with H as the data we got or more
- What part of the data space represents equal or lesser compatibility? As with the hypothesis test, we (at least implicitly) need an alternative hypothesis.



Example of *p*-value

- Sample of two radioactive isotopes with different lifetimes.
- The PDFs for the decay time will depend on the mean lifetime τ of the nucleus $f(t\,|\,\tau) = \frac{1}{\tau} e^{-\frac{t}{\tau}}$
- A nucleus is observed to decay after $t_{\text{obs}} = 0.6\ \text{s}$
- We want to calculate the *p*-value of the hypothesis H that the nucleus is from an isotope with lifetime $\tau = 0.2\ \text{s}$
- The choice of what is meant by t_{obs} less compatible with the H will depend on the alternative hypothesis (i.e. the lifetime of the other nucleus)





Interpretation of the *p*-value

- The *p*-value of H is not $P(H)$:
  - In **frequentist** statistic we don't talk about the probability of a hypothesis
- In a **Bayesian** approach we do define $p(H)$ as the degree of belief but to obtain it we need Bayes' theorem and therefore a prior $\pi(H)$

$$P(H \mid \vec{x}) = \frac{P(\vec{x} \mid H)\,\pi(H)}{\int P(\vec{x} \mid H)\,\pi(H)\,dH}$$

- In **frequentist** statistic, we have no prior, but the p-value is not to be misinterpreted as $P(H)$

The test statistic

- The *p*-value p_H can be defined from a test statistic t (the decision function). Remember that the test statistic is a scalar function of the data.

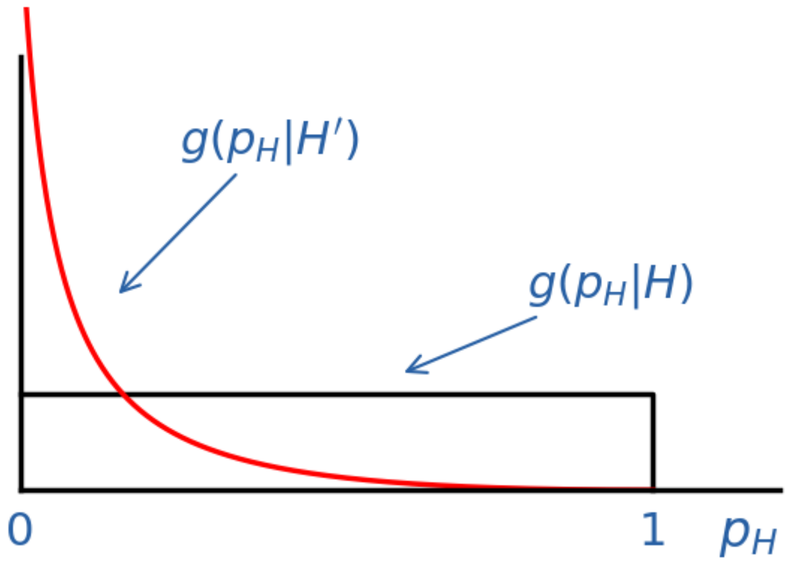
  Then the p-value for the hypothesis H is:

$$p_H = \int_{t_{\text{obs}}}^{\infty} f\left(t' \mid H\right) dt'$$

- p_H depends on the observed test statistic t_{obs} → it's a random variable
- What is the PDF of p_H under the assumption of H ?

$$g(p_H|H) = \frac{f(t|H)}{\left|\partial p_H/\partial t\right|} = \frac{f(t|H)}{f(t|H)} = 1 \qquad \left(0 \le p_H \le 1\right)$$

- Then p_H will be distributed as a uniform distribution between 0 and 1.
- Alternative hypotheses should be distributed close to 0…

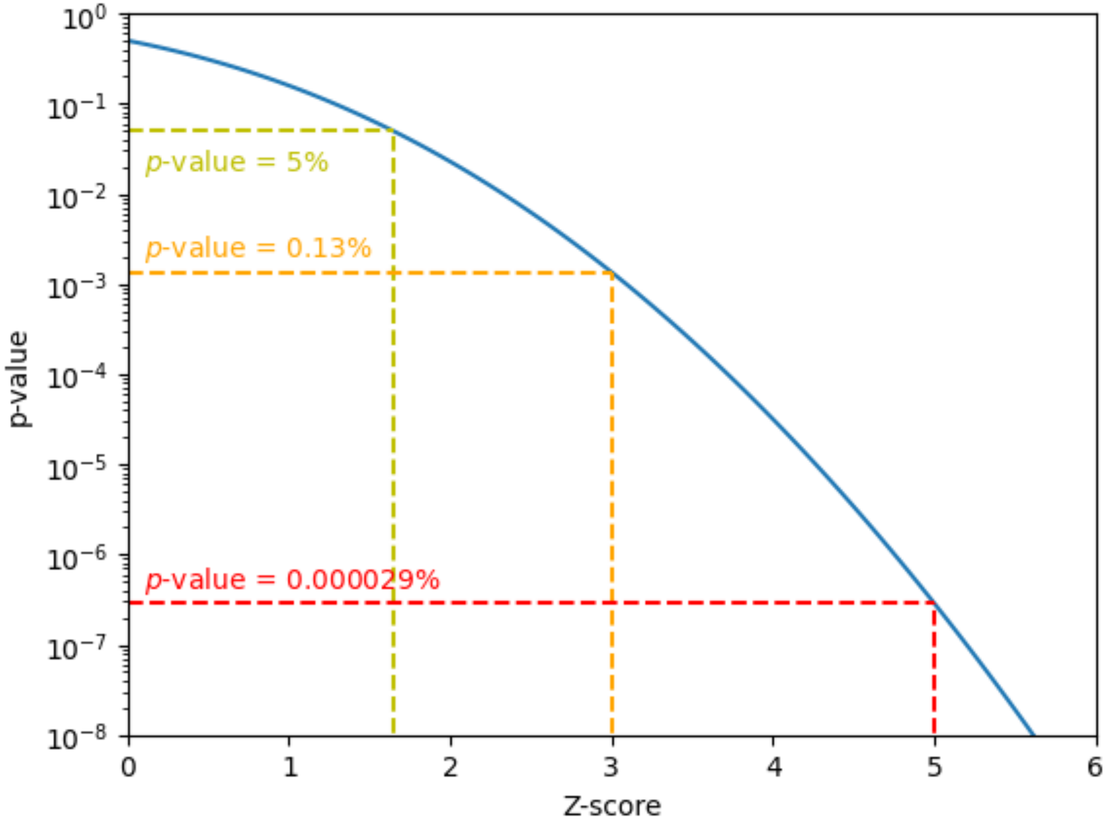


Jupyter notebook

Week2_CountingExperiment.ipynb

Discovery thresholds

- Setting a threshold for the p-value is like equivalent to defining the size α of a hypothesis test
- Thresholds for "discovery" vary greatly in the scientific literature
  - 5% is a common threshold for rejecting H_0
  - A 3 sigma threshold is sometimes considered
  - In particle physics discovery requires $Z > 5$
    - Extremely low p-value, often criticized
    - Helps to prevent effects from multiple testing (also called p-hacking or look-elsewhere effect)
    - Helps to prevent discovery claims due to unforeseen systematic effects



Frequentist parameter estimation

Main reference: G.Cowan chapter 5

Parameter estimation: definitions

- The parameters are the constants that characterise a PDF, e.g. θ in the exponential distribution:

$$f(x;\theta) = \frac{1}{\theta} e^{-x/\theta}$$

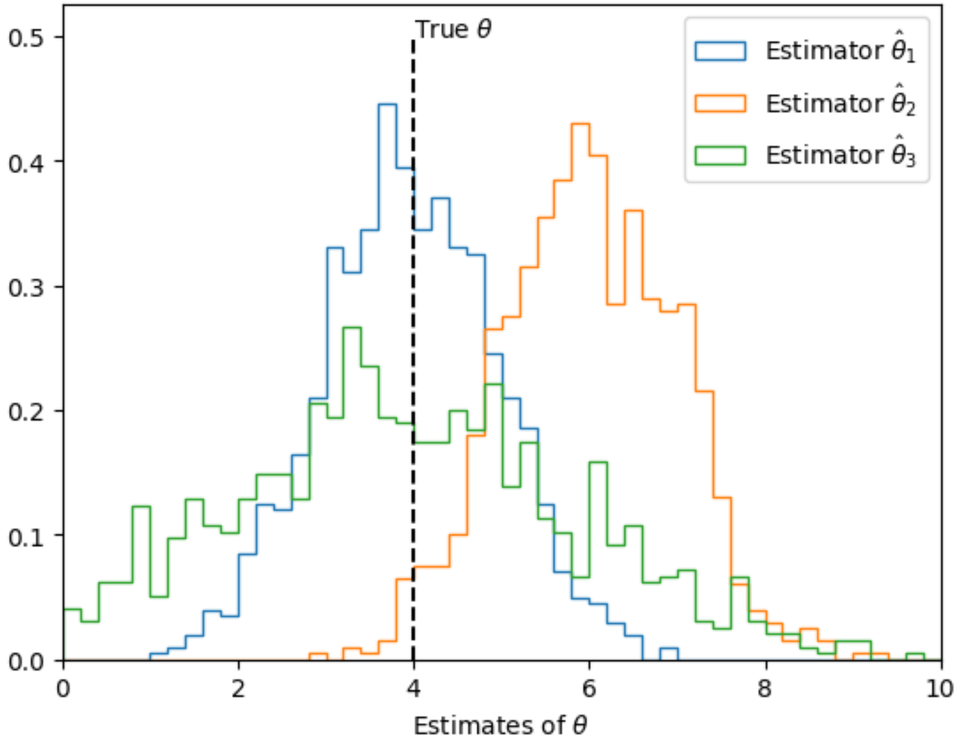
  - PDF parameters can be thought as indexing a (continuous) set of hypotheses

- An estimator is a function of the data that estimates a parameter e.g.

$\hat{\theta}(\vec{x})$ is an estimator of the parameter θ

Properties of estimators

- The estimator is a function of the data and is thus a random variable
- There is no unique way to get the “right” estimator.
- You want your estimator to have good properties:
  - We want a small bias $E[\hat{\theta}] - \theta$
  - We want a small variance $V[\hat{\theta}]$



Guess the estimator

- **Prameter**: $\mu = E[x] = \int x f(x)\, dx$ (population mean)
- **Estimator**: $\widehat{\mu} = \frac{1}{n} \sum_{i=1}^{n} x_i$ (the sample mean)
- Properties of the estimator:
  - Bias: $E[\hat{\mu}] - \mu = 0$
  - Variance: $V[\hat{\mu}] = \sigma^2/n$

Guess the estimator

- **Prameter**: $\sigma^2 = V[x] = \int (x-\mu)^2 f(x)\, dx$ (population variance)
- **Estimator**: $\widehat{\sigma^2} = \frac{1}{n-1} \sum_{i=1}^{n} \left(x_i - \bar{x}\right)^2$ (the sample variance)
- Properties of the estimator:
  - Bias: $E[\widehat{\sigma^2}] - \sigma^2 = 0$
  - Variance: $V[\widehat{\sigma^2}] = \frac{1}{n}\left(\mu_4 - \frac{n-3}{n-1}\mu_2\right)$ with μ_i being the PDF moments

The Likelihood

Main reference: G.Cowan chapter 6

The likelihood

- Suppose you have a set of measurements $\vec{x}$ that can be described by family of hypotheses modeled as a *joint* PDF $f(\vec{x};\theta)$ that depends on a parameter θ
- The **likelihood function** for a fixed dataset $\vec{x}$ is just a function of θ:

$$L(\theta) = f(\vec{x};\theta)$$

Actual definition of likelihood

- For a dataset of n ***independent**** *and identically distributed* x from a PDF $f(x;\theta)$ the likelihood is simply:

$$L(\theta) = \prod_{i=1}^{n} f(x_i;\theta)$$

The formula we'll use the most

* Remember that independence implies that $f(x_1, x_2) = f(x_1)f(x_2)$

Maximum Likelihood Estimator (MLE)

- We use the likelihood to define an estimator (no guessing)
- **Recipe**: As estimator $\hat{\theta}$ take the parameter value θ that maximizes the likelihood $L(\theta)$

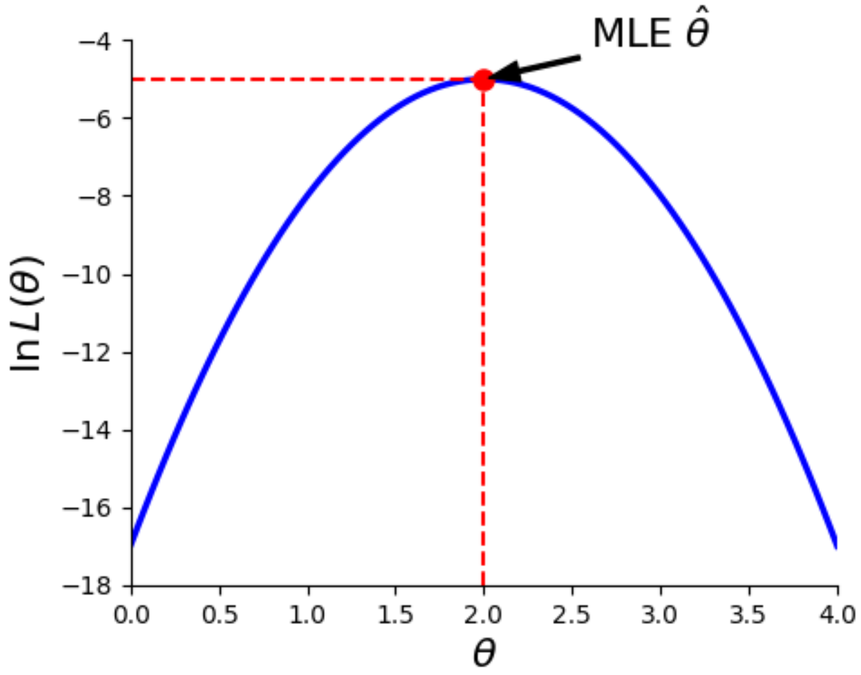
- MLE is very powerful, but not guaranteed to have optimal properties (minimal bias and variance)

- Normally we take the log of the likelihood (max at same θ)

$$\ln L(\theta) = \ln \prod_{i=1}^{n} f(x_i; \theta) = \sum_{i=1}^{n} \ln f(x_i; \theta)$$

- Often only one maximum, which is found by:

$$\frac{\partial \ln L(\theta)}{\partial \theta} = 0$$



Jupyter notebook

Week2_MLE_example.ipynb